Tao of RWD

Generating Semi-Synthetic Data for Causal Inference

The Credence Framework (*Tutorial and app*)

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Historical slides. See the repository README for the current app and export contract.

The Validation Problem

When validating causal inference methods, we often use **simulated** data:

- Tools like `simcausal` generate data with known ground truth
- We can compare naive, G-computation, IPW, and TMLE estimates
- Methods can be evaluated against the true causal effect

But how realistic is this synthetic data?

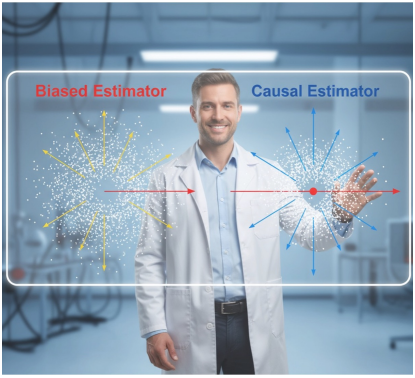
- We hand-specify the DGP with parametric assumptions
- Real-world data has complex, non-linear relationships
- Would our estimator comparisons hold in messier settings?

This tutorial: The **Credence framework** for generating more realistic synthetic data to validate causal methods

The Estimator Choice Problem

Imagine you're a researcher familiar with causal methods.

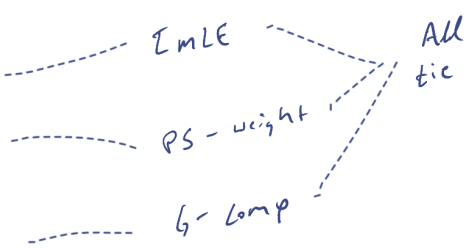
- You face a choice among estimators: IPW, AIPW/DR/DML, TMLE, matching
- How do you know which works better on messy, confounded, finite-sample data?
- If you were like me, you'd simulate 'realistic' data with `simcausal()`
- But even DAG-directed synthetic data lacks 'gritty realism'



ID	W4	W2	W3	W1	A	Y.1	Y.0	Y
1	0	0	1.2423668	0	1	1	0	1
2	0	1	1.2980802	1	0	0	1	1
3	0	1	3.4810167	0	0	0	0	0
4	0	1	1.3146951	0	0	0	0	0
5	1	1	0.5028049	1	0	0	0	0
6	0	0	1.4248857	0	0	1	0	0

Left Wanting (my experience)

ID	W4	W2	W3	W1	A	Y.1	Y.0	Y
1	0	0	1.2423668	0	1	1	0	1
2	0	1	1.2980802	1	0	0	1	1
3	0	1	3.4810167	0	0	0	0	0
4	0	1	1.3146951	0	0	0	0	0
5	1	1	0.5028049	1	0	0	0	0
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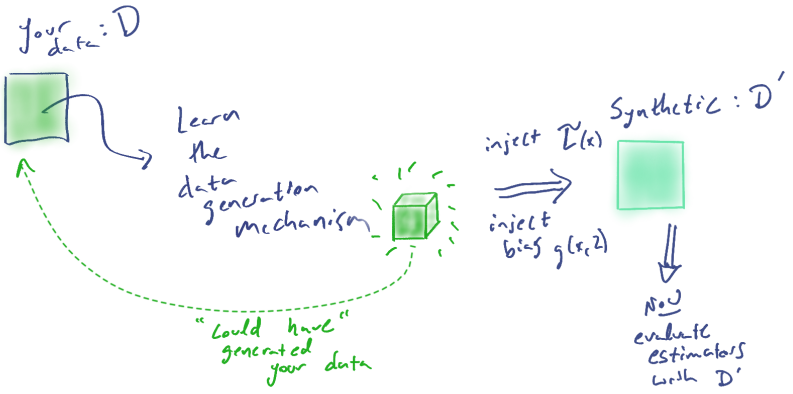
Why Traditional Simulations Fall Short

How can we create data realistic enough to choose among estimators for *real-world*, finite-sample settings?

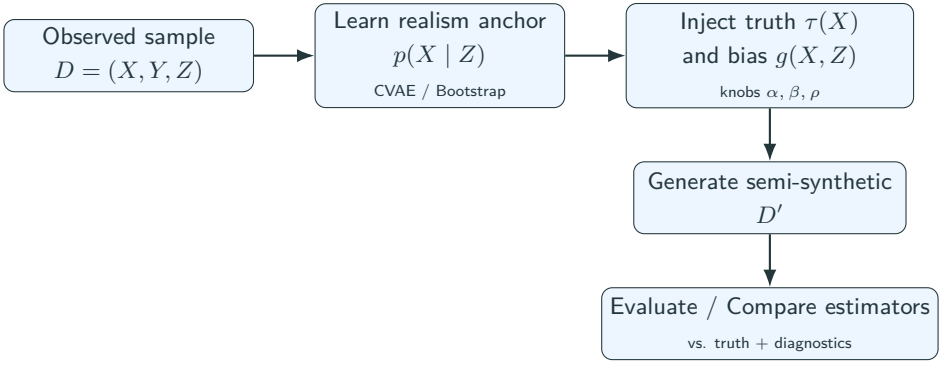
- **Embedded assumptions:** stylized DGPs can privilege certain estimators
- **Simplifications:** MVN/DAG-only sims rarely match empirical $p(X)$ and $p(X \mid Z)$
- **Entanglement:** encoding truth often breaks empirical resemblance
- **Consequence:** misspecified DGPs and weak realism can mislead

Goal: Build semi-synthetic data with known ground truth on top of realistic structure $\rightarrow$ realism first, then controlled bias injection

The Vision


your data: D
Learn the data generation mechanism
"could have" generated your data
inject $\tilde{L}(x)$
inject bias $g(x,z)$
Synthetic : D'
Now evaluate estimators with D'

The Credence Framework



realism via $p(X \mid Z)$; inject target $\tau(X)$ and bias g with rigidity (α, β); evaluate estimators under known truth.

Injecting Truth and Bias

- Choose $\tau(\mathbf{X})$ (e.g., constant or heterogeneous).
- Hidden factor $U \sim \mathcal{N}(0,1)$; baseline noise $\varepsilon \sim \mathcal{N}(0,\sigma^2)$.
- Potential outcomes: $Y(0) = h(X) + \gamma U + \varepsilon$, $Y(1) = Y(0) + \tau(X)$.
- Treatment: $\Pr(Z = 1 \mid U) = \text{logit}^{-1}(\kappa + \rho U)$ to dial unobserved confounding (κ: intercept; avoid clash with rigidity α).

Credence Realism Pipeline

Learning the realism anchor

1. Preprocess once: fixed `model.matrix` for $\mathbf{X}$; baseline $h(X)$ and propensity $\eta(X)$ for diagnostics.
2. Train CVAE to model $p(X \mid Z)$; encode (X, Z), sample latent U, decode with Z for realistic X.

Conditional ELBO (Evidence Lower BOund) balances reconstruction fidelity with KL regularization; KL keeps the latent distribution near a simple prior so we can traverse and sample the learned manifold.

Overlaying causal structure

- Inject target truth $f(x)$ and bias $g(x, z)$ with rigidity (α, β).
- Bias knobs: hidden confounding ρ, positivity trimming, measurement noise.
- Compare estimators against known truth and share diagnostics/settings.

The Credence Demo App

Python Shiny app for hands-on exploration:

- **Guided tutorial:** Walks through generating semi-synthetic data with controllable τ and ρ
- **Sandbox mode:** Experiment with alternative generators for $X \mid Z$ (Gaussian MVN, bootstrap)
- Same learn $\rightarrow$ generate $\rightarrow$ compare pipeline

Live app:

`https://andystats.shinyapps.io/causally-credible_tutorial/`

Strawman Comparison: When Do Methods Agree?

Conditions under which PS-weighting, TMLE, and DML produce similar ATE:

- **Simple DGP:** additive/linear signal; weak interactions/nonlinearity.
- **Adequate overlap:** no extreme propensities; trim tails if needed.
- **Well-specified learners:** logistic PS and linear outcome work reasonably.
- **No hidden bias:** set $\rho = 0$; measurement error off.

Expectation: all consistent methods center near the true ATE; TMLE/DR/DML offer small robustness/CI advantages.

Summary and Applications

Key takeaways:

- Credence separates realism ($p(X \mid Z)$) from causal truth ($\tau(X)$)
- Pipeline: Learn $\rightarrow$ Inject τ and bias $\rightarrow$ Compare estimators
- CVAE provides flexible realism anchors; bias knobs (ρ, positivity) enable sensitivity analysis

Applications:

- **Method development:** Test estimators under realistic conditions
- **Sensitivity analysis:** Dial bias knobs to understand robustness
- **Education:** Show when methods agree vs. diverge
- **Regulatory:** Build credible validation datasets for real-world evidence

Questions?

Try the app: `https://andystats.shinyapps.io/causally-credible_tutorial/`

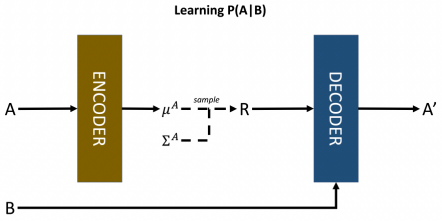
Code: `https://github.com/andystats/causally-credible`

email: wilson.stats@gmail.com

[Optional] Credence framework (expanded)

- Make $D' = (X', Y', Z')$ match D; minimize distance $d\big((X, Y, Z), (X', Y', Z')\big)$.
- **Constraints:** inject truth $f(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$ and bias $g(x, z)$ with rigidity (α, β).
- **Bias knobs:** ρ (latent confounding) and overlap/positivity; others optional.
- **Learn realism:** fit $P(X \mid Z)$ (CVAE or bootstrap); keep diagnostics lightweight.
- **Benchmark:** compare estimators against known truth and report settings.

Conditional VAE Schematic



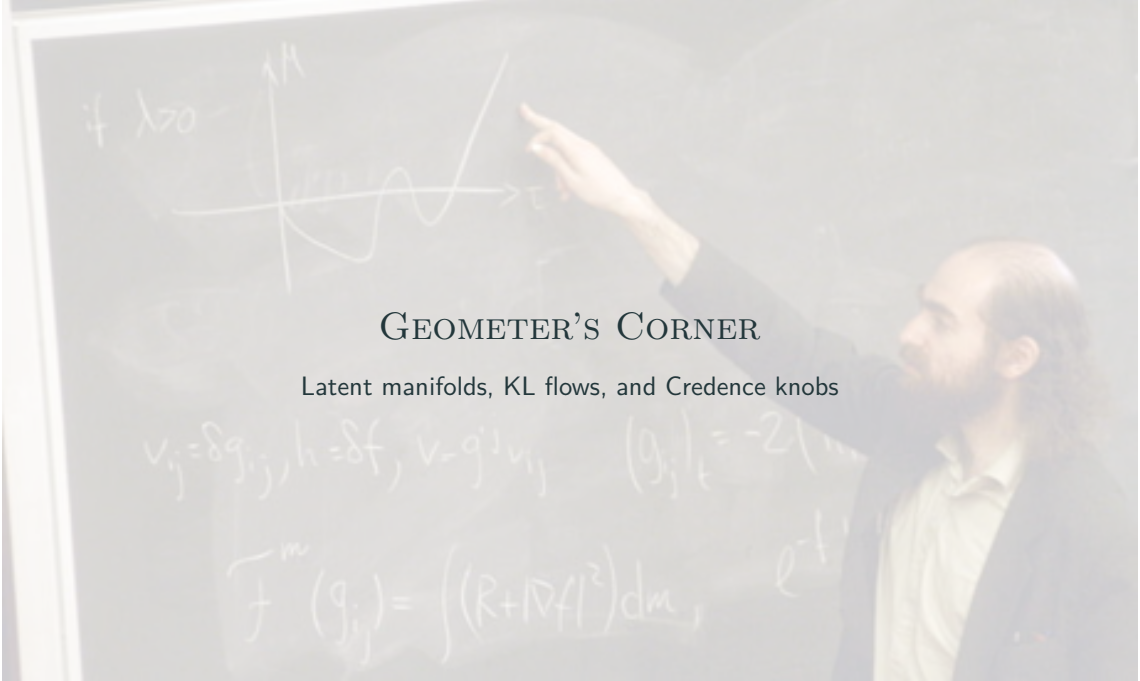
Adapted from Parikh et al. (2022).

Intuition.

- Encoder compresses (A, B) into a latent U that keeps the signal needed to rebuild A.
- Decoder mixes (U, B) to regenerate A; sampling U yields draws from $P(A \mid B)$.
- KL regularization keeps U near a simple prior so the model generalizes rather than memorizes.

Why it matters here.

- Realism knob: the CVAE learns the empirical $P(X \mid Z)$ so draws stay close to the observed covariates.
- Simulation knob: once trained, we can dial Z and sample many X quickly for Credence experiments.



Geometer's Corner

Latent manifolds, KL flows, and Credence knobs

Deep Dive: Framework (2.1) — Spaces (Figure)

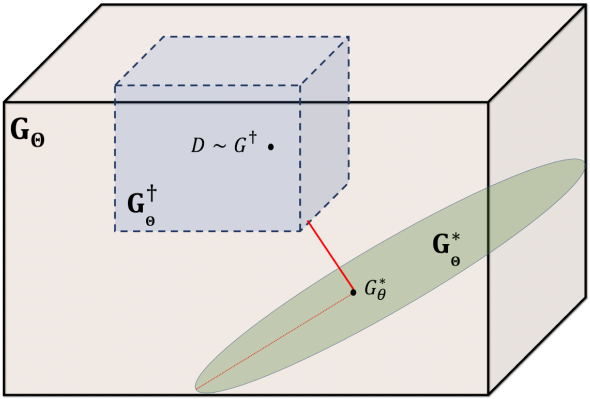
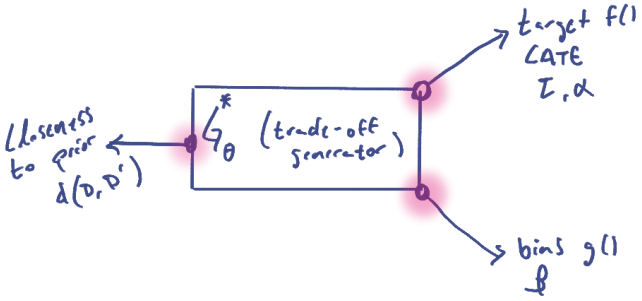


Figure 1 (Parikh et al. 2022): G_{Θ}, $G^{\dagger}$, $G^{\dagger}_{\Theta}$ (empirically isomorphic), G^{*}_{Θ}.

Goal: find G^{*}_{θ} in the overlap or closest under distance d.

Objective as Tug-of-War



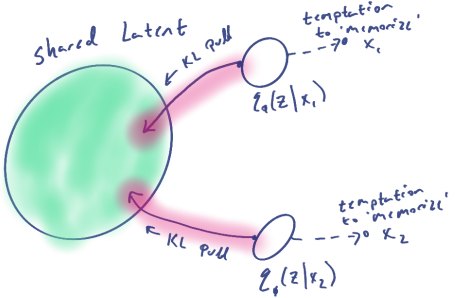
- $d(D, D')$ keeps us glued to observed manifold; α, β pull toward causal knobs
- If $\alpha = \beta = 0$, we sit in $G^{\dagger}_{\Theta} \cap G^{*}_{\Theta}$ — samples from empirical isomorphism
- KL-regularized learning explains why these forces don't tear realism apart

KL in the VAE Objective

$$\mathcal{L} = \underbrace{\mathbb{E}_{q_\phi(z|x)}\big[\log p_\theta(x \mid z)\big]}_{\text{reconstruction}} - \underbrace{D_{\mathrm{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big)}_{\text{regularization}}$$

- Encoder $q_\phi(z \mid x)$ learns latent distributions; decoder $p_\theta(x \mid z)$ maps back
- KL keeps encoders near a simple prior ($\mathcal{N}(0, I)$), shaping a usable latent manifold
- In Credence this realism anchor lets us inject truth/bias without drifting from observed geometry

Geometric Intuition: KL as Prior Pull



- KL strength acts like an attraction to the shared manifold; weak KL lets posteriors drift into memorisation spikes.
- Balancing that attraction mirrors other manifold-regularised ideas (e.g., Aleph One); Credence taps the same geometry to stay realistic.

β-VAE: Tuning the Regularization

$$\mathcal{L}_\beta = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \beta\, D_{\mathrm{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big)$$

- $\beta > 1$: stronger pull to the prior $\Rightarrow$ cleaner manifold, potential loss in reconstruction detail.
- $\beta < 1$: weaker pull $\Rightarrow$ sharper fits, risk of memorizing data and losing generative smoothness.
- Credence: treat VAE-β as the latent-space sibling of α, β, ρ; i.e., a knob that keeps realism and injected bias in balance.

Geometry Back to Credence

- **Realism manifold:** KL-regularized CVAE shapes $p(X \mid Z)$ so *'the vision'* sits on a learnable surface.
- **Knob legitimacy:** d, α, β, ρ, and VAE-β become coordinated forces, each with a geometric role.
- **Estimator insight:** when performance shifts we know if realism, truth, or bias pulled the generator off balance.
- **Full circle:** The geometry explains why Credence's knobs deliver the benchmarking datasets we wanted (and we can trust the process).

References

Credence Framework:

- Parikh H, et al. (2022). Validating Causal Inference Methods. *ICML*. arXiv:2202.04208

Related Methods:

- Kingma & Welling (2014). Auto-Encoding Variational Bayes. *ICLR*.
- Higgins et al. (2017). β-VAE. *ICLR*.

Software:

- Credence App: `https://andystats.shinyapps.io/causally-credible_tutorial/`
- `simcausal` R package